Solar System Models · Tier Pack · Independent

Learning goal: I can use solar system data and a scale model to link distance from the Sun to orbit time.

Independent Route (Book First)

Use the data table to state the pattern linking distance from the Sun to orbit time (quoting two planets' numbers), and explain one strength and one limitation of the strip model.

Help: As distance from the Sun increases from __ (Mercury) to __ (Neptune), orbit time increases from __ to __. One limitation of the strip model is __.

Optional Make It · Independent

Make a 100 cm (or folded 20 cm) Scaled Solar System Strip

Mark the scaled planet positions from the Sun (0 cm) to Neptune, label the orbit times for Earth and Neptune, and write your conclusion in your book.

Word Help

orbit time: the time a planet takes to travel once all the way around the Sun (its year)

scale model: a model that shrinks huge real distances or sizes using a fixed rule

astronomical data: measurements of planets such as distance from the Sun, diameter, and orbit time

limitation: something a science model leaves out or

cannot show accurately

Exit Check

Which planet takes longer to orbit the Sun: Earth (365 days) or Neptune (165 years)?

State the pattern linking a planet's distance from the Sun to its orbit time.